

Software Engineering Mastery Roadmap

Fully Revised Edition with Embedded Self-Testing

Stage 1 — Python Rebuild

Core Topics: Variables and types, Control flow, Functions, Lists/Tuples/Dictionaries/Sets, Files, Exceptions, Modules

Embedded Self-Test Checklist

- Can explain from memory
- Can write code from a blank file
- Can choose correct data structure
- Can debug errors without AI
- Can build a small project independently

Mastery Requirement: Do not mark this stage complete until you can explain, implement, debug, modify, and apply the concepts to an unfamiliar problem.

Stage 2 — Problem Solving

Core Topics: Counting, Searching, Filtering, Recursion, Parsing

Embedded Self-Test Checklist

- Can derive solution before coding
- Can write pseudocode
- Can test edge cases
- Can explain complexity

Mastery Requirement: Do not mark this stage complete until you can explain, implement, debug, modify, and apply the concepts to an unfamiliar problem.

Stage 3 — Data Structures

Core Topics: Arrays, Linked Lists, Stacks, Queues, Hash Tables, Trees, Graphs

Embedded Self-Test Checklist

- Can implement from scratch
- Can explain trade-offs
- Can explain memory representation
- Can choose appropriate structure

Mastery Requirement: Do not mark this stage complete until you can explain, implement, debug, modify, and apply the concepts to an unfamiliar problem.

Stage 4 — Algorithms

Core Topics: Search, Sorting, DFS, BFS, Greedy, Dynamic Programming

Embedded Self-Test Checklist

- Can implement from memory
- Can explain correctness
- Can compare alternatives
- Can justify complexity

Mastery Requirement: Do not mark this stage complete until you can explain, implement, debug, modify, and apply the concepts to an unfamiliar problem.

Stage 5 — C

Core Topics: Pointers, Memory, Structs, Heap/Stack, Compilation

Embedded Self-Test Checklist

- Can draw memory diagrams
- Can debug memory issues
- Can build structures without assistance

Mastery Requirement: Do not mark this stage complete until you can explain, implement, debug, modify, and apply the concepts to an unfamiliar problem.

Stage 6 — C++

Core Topics: OOP, Templates, STL, RAII, Smart Pointers

Embedded Self-Test Checklist

- Can design classes
- Can explain ownership
- Can use STL appropriately

Mastery Requirement: Do not mark this stage complete until you can explain, implement, debug, modify, and apply the concepts to an unfamiliar problem.

Stage 7 — Git & Linux

Core Topics: Git workflow, Branching, Merging, Linux CLI

Embedded Self-Test Checklist

- Can complete project with Git
- Can resolve merge conflicts
- Can work from terminal

Mastery Requirement: Do not mark this stage complete until you can explain, implement, debug, modify, and apply the concepts to an unfamiliar problem.

Stage 8 — Operating Systems

Core Topics: Processes, Threads, Virtual Memory, Concurrency

Embedded Self-Test Checklist

- Can explain program execution lifecycle
- Can explain race conditions and deadlocks

Mastery Requirement: Do not mark this stage complete until you can explain, implement, debug, modify, and apply the concepts to an unfamiliar problem.

Stage 9 — Networking

Core Topics: TCP/IP, DNS, HTTP, Sockets

Embedded Self-Test Checklist

- Can explain browser-to-server request flow

Mastery Requirement: Do not mark this stage complete until you can explain, implement, debug, modify, and apply the concepts to an unfamiliar problem.

Stage 10 — Databases

Core Topics: SQL, Indexes, Transactions, Normalization

Embedded Self-Test Checklist

- Can design schema
- Can write queries
- Can explain ACID

Mastery Requirement: Do not mark this stage complete until you can explain, implement, debug, modify, and apply the concepts to an unfamiliar problem.

Stage 11 — Backend APIs

Core Topics: REST, Authentication, Validation, Persistence

Embedded Self-Test Checklist

- Can build API independently
- Can test endpoints

Mastery Requirement: Do not mark this stage complete until you can explain, implement, debug, modify, and apply the concepts to an unfamiliar problem.

Stage 12 — Testing & Debugging

Core Topics: Unit Tests, Integration Tests, Debuggers, Profiling

Embedded Self-Test Checklist

- Can write meaningful tests
- Can reproduce and isolate bugs

Mastery Requirement: Do not mark this stage complete until you can explain, implement, debug, modify, and apply the concepts to an unfamiliar problem.

Stage 13 — System Design

Core Topics: Scalability, Caching, Queues, Reliability

Embedded Self-Test Checklist

- Can explain architecture trade-offs

Mastery Requirement: Do not mark this stage complete until you can explain, implement, debug, modify, and apply the concepts to an unfamiliar problem.

Stage 14 — Docker & Cloud

Core Topics: Containers, Images, Compose, Deployment

Embedded Self-Test Checklist

- Can containerize and deploy a project

Mastery Requirement: Do not mark this stage complete until you can explain, implement, debug, modify, and apply the concepts to an unfamiliar problem.

Stage 15 — Portfolio Projects

Core Topics: CLI, Backend, Systems, Capstone

Embedded Self-Test Checklist

- Can complete and document projects independently

Mastery Requirement: Do not mark this stage complete until you can explain, implement, debug, modify, and apply the concepts to an unfamiliar problem.

Stage 16 — Job Readiness

Core Topics: Interviews, GitHub, Communication

Embedded Self-Test Checklist

- Can explain projects and engineering decisions

Mastery Requirement: Do not mark this stage complete until you can explain, implement, debug, modify, and apply the concepts to an unfamiliar problem.

Stage 17 — Specialization

Core Topics: Backend, Systems, Cloud, ML, etc.

Embedded Self-Test Checklist

- Can apply fundamentals in chosen domain

Mastery Requirement: Do not mark this stage complete until you can explain, implement, debug, modify, and apply the concepts to an unfamiliar problem.

Universal Self-Testing Rules

- Can I explain it without notes?
- Can I implement it from memory?
- Can I debug a broken version?
- Can I modify it for a new requirement?
- Can I apply it to an unfamiliar problem?
- Can I teach it to another person?